

# DermAI: Multimodal Skin Disease Detection & Dermatological Consultation Platform

An end-to-end clinical assistive system combining deep learning computer vision (ResNet-50 / EfficientNet), Explainable AI (Grad-CAM heatmaps), and Multilingual RAG Conversational AI for early lesion screening.

**Author:** Mohd Hammad

**Domain:** Deep Learning & Healthcare AI

**Datasets:** HAM10000 + ISIC 2019

**Tech Stack:** React + Node.js + FastAPI + PyTorch + Gemini

## 1 Project Problem Statement & Core Motivation (Kyun Banaya?)

Skin disease is one of the most common human illnesses globally, affecting over 900 million people at any given time. However, healthcare accessibility presents three severe bottlenecks:

### Critical Doctor Shortage

In developing regions and Tier-2/3 cities, the dermatologist-to-population ratio is less than **1 per 100,000 citizens**. Patients travel hundreds of kilometers for routine skin consultations.

### Fatal Diagnostic Delays

Malignant Melanoma caught in Stage I boasts a **99% 5-year survival rate**. When detected late after metastasis, survival plunges below **30%**. Early screening saves lives.

### Dangerous Self-Medication

Patients search uncured web forums and apply OTC steroid creams without knowing if a lesion is fungal, bacterial, or cancerous, causing irreversible tissue aggravation.

## 2 High-Level Architecture Overview (Konsi Cheez Kaise Kaam Karti Hai?)

DermAI operates as a distributed **Multimodal Microservices Architecture** where each layer is decoupled for scalability, high throughput, and safety:

```
1. CLIENT LAYER (React 18 + Vite + Tailwind CSS) [Port 5173] |
| • Drag & Drop Image Upload • Real-time Grad-CAM Heatmap Slider Toggle |
| • Bilingual Chat UI (English & Hindi) • OTP-protected Password Reset & Doctor Map |
|
| HTTP / REST (JWT Auth Token)
|
2. API GATEWAY & BUSINESS CORE (Node.js + Express.js) [Port 5000] |
| • Rate Limiting & Helmet Security • BCrypt Password Hashing & 6-digit OTP Verification |
| • Clinical Safety Filter Layer • Google Gemini Multilingual Conversational Routing |
|
| Multipart/Form-Data Image | Coordinates Query
|
3. DEEP LEARNING ENGINE (FastAPI + PyTorch) | 4. GEOLOCATION DOCTOR FINDER |
| [Port 8000] | | • OpenStreetMap Nominatim API |
| • ResNet-50 / EfficientNet-B0 Backbone | | • Overpass Dermatologist Query |
| • 7-Class HAM10000 + ISIC Probabilities | | • Local Verified Doctor DB |
```

• Grad-CAM Visual Heatmap Generation |

### 3 Deep Dive: Core Components & Technical Rationale (Kyu Use Kiya?)

#### A. Deep Learning Vision Engine

FastAPI + PyTorch

**Kaise Kaam Karta Hai:** Uploaded dermoscopic photo 224x224 RGB tensor mein convert hoti hai, ImageNet mean/std se normalize hoti hai, aur deep convolutional backbone se pass hoti hai. Final dense layer 7 disease classes ke softmax probability distribution calculate karti hai.

**Kyu Use Kiya:** Python/FastAPI Asynchronous C-based tensor operations support karta hai jo Node.js mein slow hote hain. PyTorch industry standard model training aur dynamic gradient computation facilitate karta hai.

#### B. Explainable AI: Grad-CAM

Visual Interpretability

**Kaise Kaam Karta Hai:** Target predicted class score ke gradients compute karta hai with respect to last convolutional feature maps (Layer4). In gradients ko spatial features se multiply karke 2D heatmap banta hai jo image ke key visual patterns ko highlight karta hai.

**Kyu Use Kiya:** AI medicine mein "Black-Box" models par doctors trust nahi karte. Grad-CAM prove karta hai ki AI ne actual skin lesion/pigment border dekhkar diagnose kiya hai, na ki ruler ya background hair dekhkar!

#### C. Multilingual Conversational AI

Gemini + Custom Safety

**Kaise Kaam Karta Hai:** Dynamic Language Matching system: Agar user English mein baat kare to English default rehti hai; agar user Hindi/Hinglish mein puche (e.g. "kya yeh dhoop se hua hai?"), to LLM automatically natural Hindi/Hinglish mein medical guidance deta hai.

**Kyu Use Kiya:** Bharat ke har patient ko English fluent nahi hoti. Bilingual conversational AI se rural aur semi-urban users bhi apni bhasha mein consultation le sakte hain bina kisi intimidation ke.

#### D. Clinical Safety Guardrail Layer

RAG + Zero-Hallucination

**Kaise Kaam Karta Hai:** Har response se pehle Regex aur Semantic Safety checks run hote hain. Agar LLM dangerous oral prescription drugs (jaise oral steroids, immunosuppressants, antibiotics) recommend kare, to filter block karke safe OTC hygiene aur dermatologist visit suggest karta hai.

**Kyu Use Kiya:** Medical ethics! Unverified prescription medicine patient ki jaan khatre mein daal sakti hai. AI ka kaam assist karna hai, certified doctor ki jagah lena nahi.

### 4 Datasets & The 7 Clinical Diagnostic Classes

DermAI ko 2 globally recognized clinical benchmark datasets par train aur evaluate kiya gaya hai:

**1. HAM10000 Dataset:** 10,015 dermoscopic images (ViDIR Group, Austria & Queensland, Australia). Gold standard benchmark for pigmented skin lesion analysis.

**2. ISIC 2019 Challenge Dataset:** 25,331 clinical images spanning broader demographic cohorts, skin tones, and rare dermatological pathologies.

| Code | Condition Name | Clinical Description & Symptoms                                                                                     | Risk Tier |
|------|----------------|---------------------------------------------------------------------------------------------------------------------|-----------|
| mel  | Melanoma       | Most dangerous malignant skin cancer arising from melanocytes; irregular borders, color variation, fast metastasis. | High Risk |

| Code  | Condition Name       | Clinical Description & Symptoms                                                                               | Risk Tier       |
|-------|----------------------|---------------------------------------------------------------------------------------------------------------|-----------------|
| bcc   | Basal Cell Carcinoma | Most common skin cancer; pearly, waxy bump or scar-like lesion resulting from chronic cumulative UV exposure. | Moderate-High   |
| akiec | Actinic Keratoses    | Precancerous scaly/crusty sun patches; small percentage progresses to Squamous Cell Carcinoma if untreated.   | Precancerous    |
| bkl   | Benign Keratosis     | Non-cancerous seborrheic keratoses, solar lentigines, or lichenoid keratoses commonly seen in aging skin.     | Benign / Low    |
| nv    | Melanocytic Nevi     | Common benign moles and birthmarks; uniform coloration with symmetrical borders.                              | Benign / Normal |
| vasc  | Vascular Lesions     | Cherry angiomas, pyogenic granulomas, and vascular malformations originating from blood vessels.              | Benign / Low    |
| df    | Dermatofibroma       | Benign fibrous skin nodules, often developing on lower legs following minor trauma or insect bites.           | Benign / Low    |

## 5 Project Fayde vs Nuksaan (Comprehensive Advantages & Limitations)

Ek strong engineering evaluation mein model ke strengths aur genuine constraints dono ka clear understanding hona chahiye:

### ☀️ Fayde (Key Advantages & Innovations)

- ✓ **Ultra-Fast Early Screening (<2 sec):** Patient ko doctor ke appointment ke liye hafte wait nahi karna padta; turant risk level ka primary estimate mil jata hai.
- ✓ **Explainable AI Trust (Grad-CAM):** Visual heatmaps se doctor aur examiner dono dekh sakte hain ki prediction random guess nahi, balki specific pathological regions par based hai.
- ✓ **Bilingual Accessibility (Hindi + English):** Dynamic language adaptation se Hindi/Hinglish bolne wale aam log bhi apni zubaan mein natural conversation kar sakte hain.
- ✓ **Strict Safety Guardrails:** Self-medication ka sabse bada khatra khatam ho jata hai kyunki system kisi bhi dangerous prescription steroid ko prescribe nahi karta.
- ✓ **Seamless Offline Viva / Presentation Mode:** Internet SMTP ya SMS gateway fail hone par bhi system 6-digit verification code screen par auto-fill kar deta hai, ensuring 100% smooth viva demo.
- ✓ **End-to-End Patient Journey:** Bas prediction dekar chhodta nahi—nearby certified dermatologists dhoondh kar Google Maps direction deta hai.

### ⚠️ Nuksaan & Limitations (Technical Constraints)

- ⚠️ **Not a Certified Diagnostic Replacement:** DermAI ek decision-support assistive tool hai. Final diagnosis ke liye physical dermatoscope aur biopsy/histopathology anivarya hai.
- ⚠️ **Image Quality & Lighting Dependency:** Smartphone camera se khinchee hui blurry, low-light, ya flash-glare wali photos par prediction confidence drop ho sakti hai.
- ⚠️ **Fitzpatrick Skin Tone Imbalance:** Public datasets (HAM10000) mein Type I-III (fair skin) ka proportion zyada hai aur Type V-VI (darker skin) kam hai, jisse domain shift ho sakta hai.
- ⚠️ **Telecom SMS Gateway Costs:** Real SMS phone par bhejne ke liye TRAI DLT commercial registration aur paid API credits lagte hain (isliye Email SMTP + On-Screen Demo fallback use kiya gaya).
- ⚠️ **Edge-Case Overlaps:** Early-stage Melanoma aur atypical Nevi (moles) visually 90% identical lagte hain, jahan model occasionally higher uncertainty flag karta hai.

## 6 Security & Authentication Architecture

### BCrypt Password Hashing

User passwords database mein plaintext save nahi hote. 12-round salt work factor ke sath cryptographically hash hote hain, securing against rainbow-table attacks.

### Stateless JWT Auth

Client authentication JSON Web Tokens (JWT) se manage hoti hai jo 7-day validity ke sath secure headers mein travel karti hai, reducing database lookups.

### 6-Digit Expiring OTP Flow

Password reset ke liye crypto-secure random 6-digit OTP generate hota hai jo 5-minute memory store mein expire hota hai, brute-force security ke sath.

## 7 Doctor Finder Integration (OpenStreetMap & Nominatim)

Patient ko AI risk assessment milne ke baad real medical consultation dilana sabse critical step hai:

- **Reverse Geocoding:** User ke browser location (Latitude/Longitude) ko OpenStreetMap *Nominatim API* se clean city/area name mein convert kiya jata hai.

- **Overpass API Query:** Nearby registered dermatologists, skin clinics, aur hospitals ko spatial radius (5km – 15km) mein query karta hai.
- **Instant Local Fallback:** Agar network connection slow ho ya API rate limit hit ho, to system instant Oms pre-seeded verified specialist database se certified doctors display karta hai, complete with approximate consultation fee, address, contact, and 1-click Google Maps navigation.

## 8 Top Viva Questions & Crisp Answers (Examiners Ke Favorite Sawaal)

Viva mein examiners ke saamne confident, technical aur precise bolne ke liye ye key questions yaad rakhein:

### Q1 Aapke model ne medical black-box problem ko kaise solve kiya?

**Answer:** Humne **Grad-CAM (Gradient-weighted Class Activation Mapping)** integrate kiya hai. Grad-CAM final convolutional layer ke feature maps par target class ke gradients compute karke ek 2D color heatmap generate karta hai. Isse doctor ko visually dikhta hai ki model ne skin ke actual lesion borders aur pigments ko analyze kiya hai, kisi background artifact ko nahi.

### Q2 Aapne dono HAM10000 aur ISIC 2019 datasets kyu use kiye, sirf ek kyu nahi?

**Answer:** HAM10000 ek high-quality curated dataset hai (10,015 images), lekin wo primary Taur par Australian aur Austrian clinics ki demographic population se aati hai. **ISIC 2019 (25,331 images)** add karne se training distribution diverse hoti hai, overfitting kam hoti hai, aur broader skin types aur rare lesion categories par model ka generalization significantly improve hota hai.

### Q3 LLM hallucinations ko rokne ke liye kya mechanism banaya hai?

**Answer:** Humne dual-layer safety architecture lagaya hai: Pehla, **Grounded Medical RAG** jo context ko clinically verified skin facts tak restrict karta hai. Doosra, hamara custom **Safety Service Filter** jo response generate hone ke baad real-time parse karta hai—agar model kisi dangerous oral prescription drug (jaise systemic steroids ya antibiotics) ka naam le, to use instantly filter karke safe OTC hygiene aur physical doctor visit advice se replace kar deta hai.

### Q4 Melanoma ko sabse high-priority alert kyu diya gaya hai?

**Answer:** Melanoma skin cancer cases ka sirf ~1% hota hai, lekin skin cancer deaths ka **75% se zyada** cause karta hai kyunki iske cancerous melanocytes lymph nodes aur internal organs mein bahut fast metastasize (spread) hote hain. Isliye Melanoma detect hote hi system highest emergency visual alert trigger karta hai.

### Q5 Kya yeh system certified dermatologist ko replace kar sakta hai?

**Answer:** Bilkul nahi. DermAI ek *Clinical Decision Support System (CDSS)* hai jiska maqsad early screening aur triage facilitate karna hai, na ki definitive biopsy diagnose karna. Hamara platform explicitly medical disclaimer ke sath patient ko qualified dermatologist tak navigate karta hai.

### Q6 Future scope aur production scaling ke liye kya roadmap hai?

**Answer:** Future roadmap mein teen main enhancements hain: (1) **Mobile App (React Native)** with on-device camera guidance to avoid blurry pictures, (2) **TensorRT / ONNX Runtime** lightweight quantized model deployment taaki phone ke andar bina internet ke bhi inference run ho sake, aur (3) Hospital EHR (Electronic Health Records) integration via standard HL7/FHIR protocols.